Sales Analysis Use case Insights and Answers

1) Which product id giving the highest profit?

We can answer this question using 2 methods

A) From the table visual in **Products** page, we can find that the most profitable product as shown in the visual is Apple smart phone, Full size which has the product_id = TEC-PH-3184

Least 5 | Top 5 | ALL

Top 5 Products	Sub-Category	Category	Units Sold	Units Sold by Month	Delivery	Sales	Discounts	COG	Net Profit	Profit Margin
Apple Smart Phone, Full Size	Phones	Technology	171		3.9	$86,921	$6,282	$21,450	$65,471	75%
Canon imageCLASS 2200 Advanced Copier	Copiers	Technology	20		4.0	$61,595	$5,599	$7,751	$53,844	87%
Cisco Smart Phone, Full Size	Phones	Technology	139		4.0	$76,424	$2,042	$14,133	$62,291	82%
Motorola Smart Phone, Full Size	Phones	Technology	134		4.2	$73,132	$5,189	$20,263	$52,869	72%
Nokia Smart Phone, Full Size	Phones	Technology	147		3.6	$71,884	$6,006	$20,596	$51,288	71%
Total			**611**		**3.9**	**$369,956**	**$25,118**	**$84,193**	**$285,763**	**77%**

B) From the DAX Measure called **Top Profit Product ID** it will give us the product_id = TEC-PH-3184

```
Top Profit Product ID =
VAR TopProduct =
    TOPN(1,
        SUMMARIZE('Fact_Sales', 'Dim_Product'[Product ID], "Total_Profit", SUM(Fact_Sales[Net Profit])),
        [Total_Profit],
        DESC
    )
RETURN
    CONCATENATEX(TopProduct, Dim_Product[Product ID], ", ")
```

2) Which product id the lowest sales amount?

We can answer this question using 2 methods

A) From the table visual in **Products** page, we can find that the least sellable product as shown in the visual is Eureka Disposable Bags for Sanitaire Vibra Groomer I Upright Vac which has the product_id = OFF-AP-4215

Least 5 | Top 5 | ALL

Least 5 Products	Sub-Category	Category	Units Sold	Units Sold by Month	Delivery	Sales	Discounts	COG	Net Profit	Profit Margin
Eureka Disposable Bags for Sanitaire Vibra Groomer I Upright Vac	Appliances	Office Supplies	2		4.0	$1	$1	$49	($48)	-4780%
Avery 5	Labels	Office Supplies	2		2.0	$5	$0	$108	($103)	-2060%
Avery Hi-Liter Comfort Grip Fluorescent Highlighter, Yellow Ink	Art	Office Supplies	4		6.0	$6	$0	$156	($150)	-2500%
Total			**20**		**4.3**	**$46**	**$4**	**$832**	**($786)**	**-1710%**

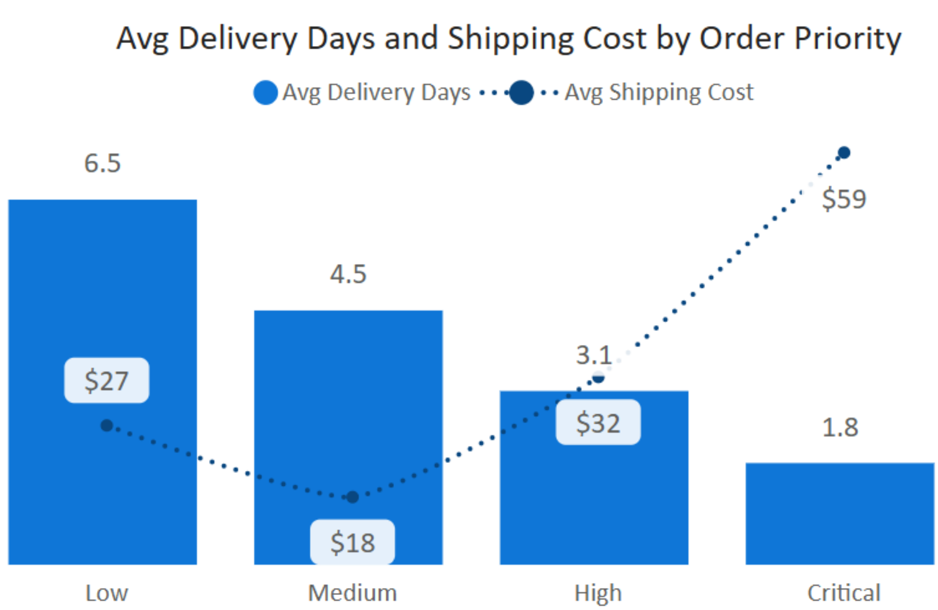
B) From the DAX Measure called **Least Sales Product ID** it will give us the product_id = OFF-AP-4215

```
Lowest Sales Product ID =
VAR TopProduct =
    TOPN(1,
        SUMMARIZE('Fact_Sales', 'Dim_Product'[Product ID], "Total_Sales", SUM(Fact_Sales[Sales Amount])),
        [Total_Sales],
        ASC
    )
RETURN
    CONCATENATEX(TopProduct, Dim_Product[Product ID], ", ")
```

3) What are the average delivery days per order priority?

From the attached column chart visual we can answer this question as following: -

- For **Low** order priority it's 6.5 days
- For **Medium** order priority it's 4.5 days
- For **High** order priority it's 3.1 days
- For **Critical** order priority it's 1.8 days



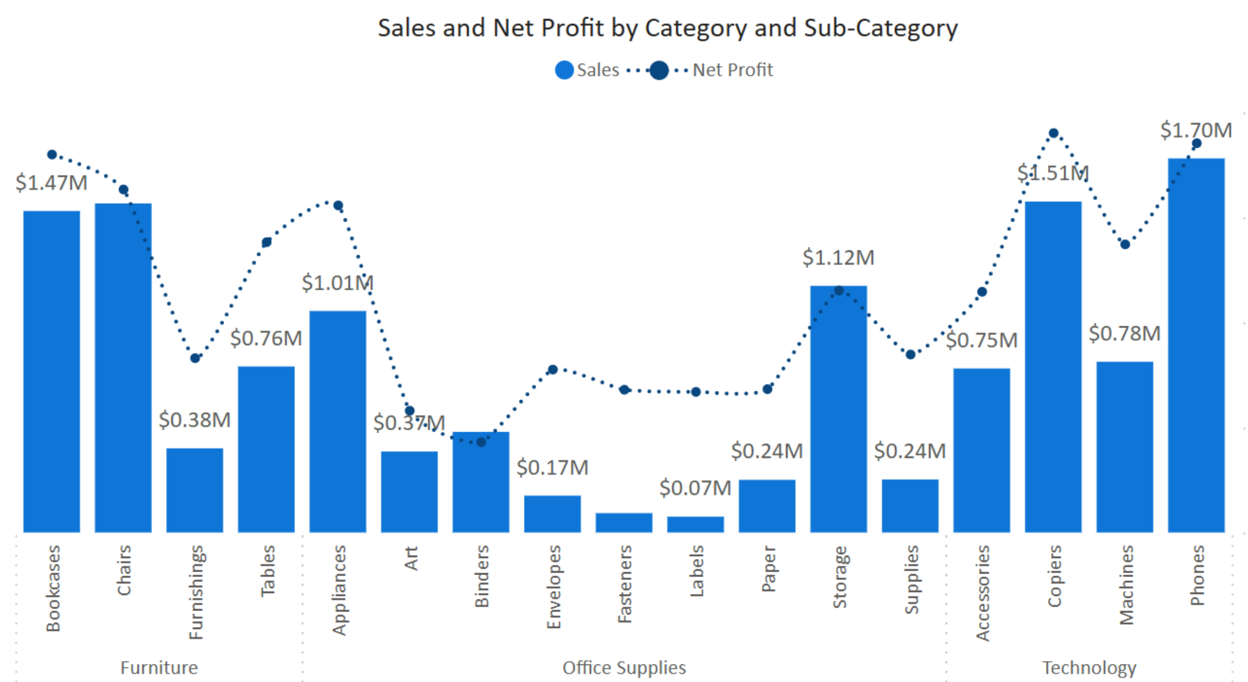
4) Which Sub-category is the highest in both Sales and Profit together?

From the attached visual we can see that **Phones** is the highest sub-category in both the sales and profit.

Top Sub-Category: Phones

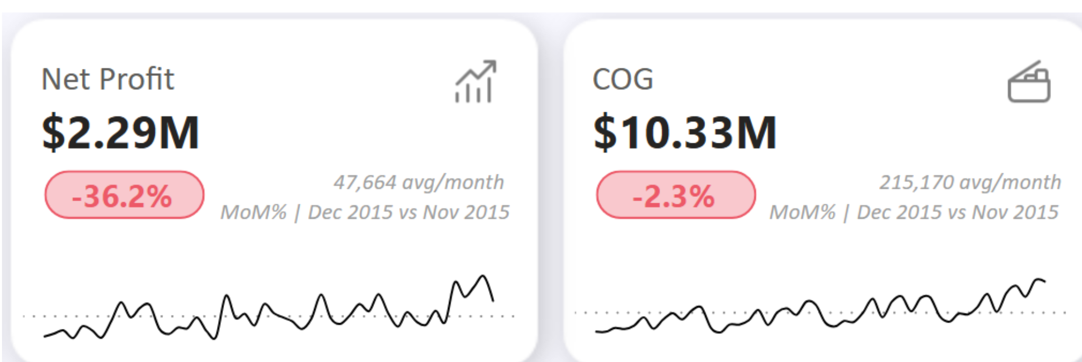
Total Sales: $1,704,935

Total Profit: $853,082.67



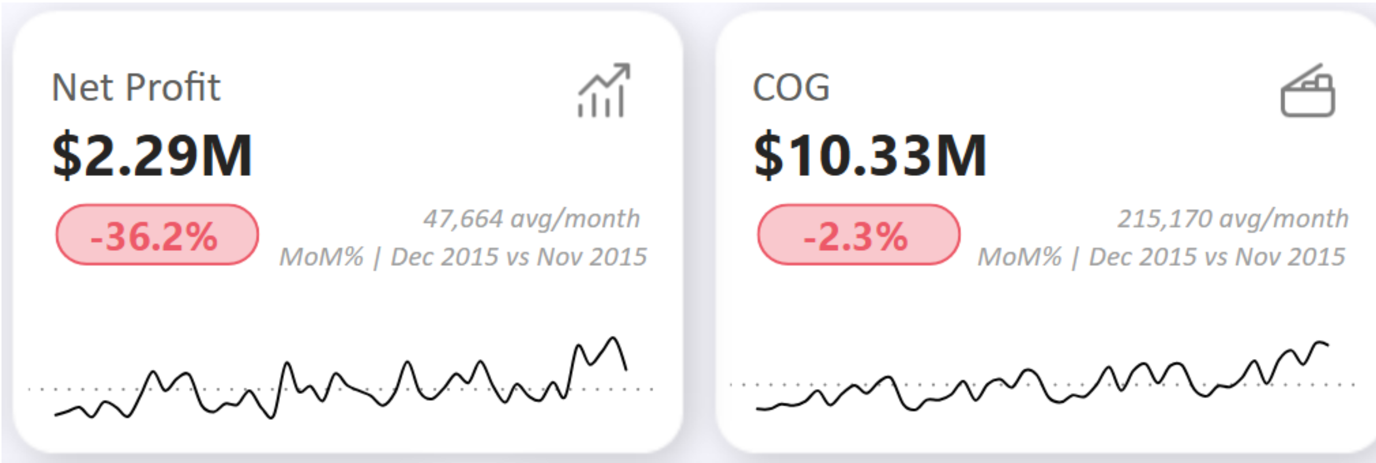
5) What is the total company net profit?

Total Company Net Profit: $2,287,883 ≈ $2.29M



6) What is the total company cost?

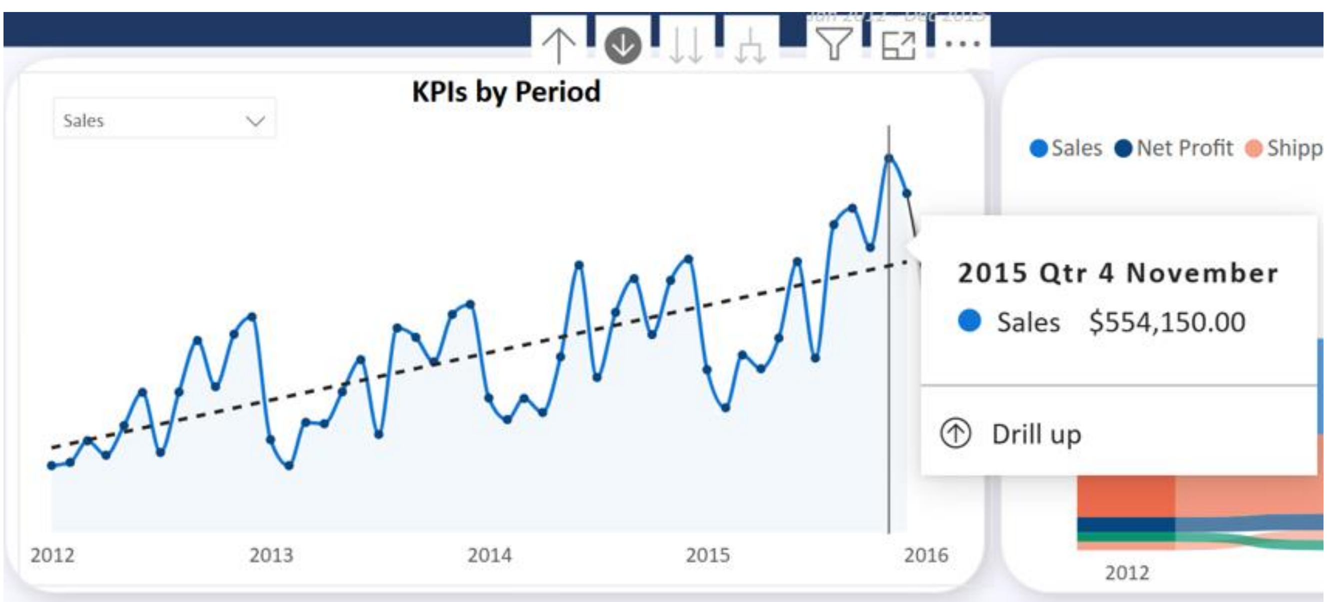
Total Company Cost: $10,328,138 ≈ $10.33M



7) What is the best month for Sales during the year (2015)?

We can answer this questions in 2 ways

A) Using this line chart in the **Time analysis** page we can find that **November** is the best month



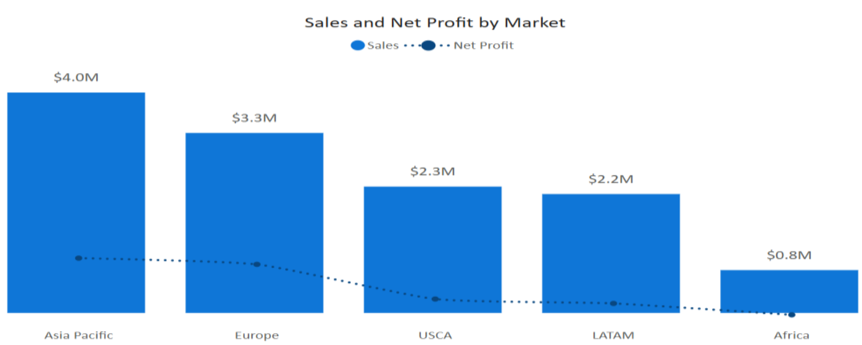
B) Using the DAX measure called **[Top Sales Month 2015]()**

```
Top Sales Month 2015 =
VAR BestMonth =
    TOPN(1,
        SUMMARIZE(
            FILTER('Fact_Sales', Fact_Sales[order year] = 2015),
            'Fact_Sales'[order month],
            "Total_Sales", [Sales]
        ),
        [Total_Sales],
        DESC
    )
RETURN
    CONCATENATEX(BestMonth, Fact_Sales[order month], ", ")
```

8) What is the most profitable Market?

We can answer this questions in 2 ways

A) Using this line chart in the **Geography** page we can find that **[Asia Pacific]()** is the best month



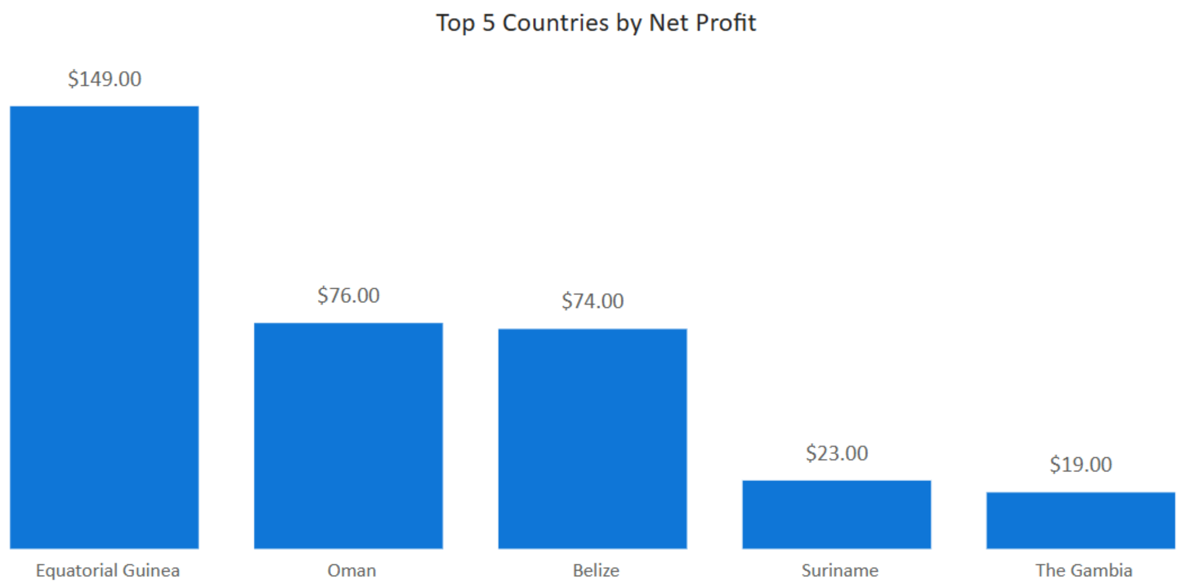
B) Using the DAX measure called **[Top Profit Market]()**

```
Top Profit Market =
VAR TopProduct =
    TOPN(1,
        SUMMARIZE('Fact_Sales', 'Dim_Geography'[Market], "Total_Profit", SUM(Fact_Sales[Net Profit])),
        [Total_Profit],
        DESC
    )
RETURN
    CONCATENATEX(TopProduct, Dim_Geography[Market], ", ")
```

9) What is the least sales country?

We can answer this questions in 2 ways

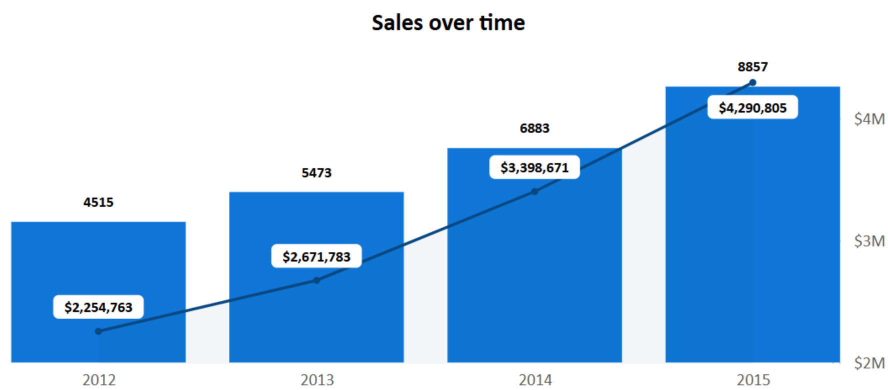
A) Using this line chart in the **geography** page we can find that **The Gambia** is the least sales country.



B) Using the DAX called **Least Country Sales**

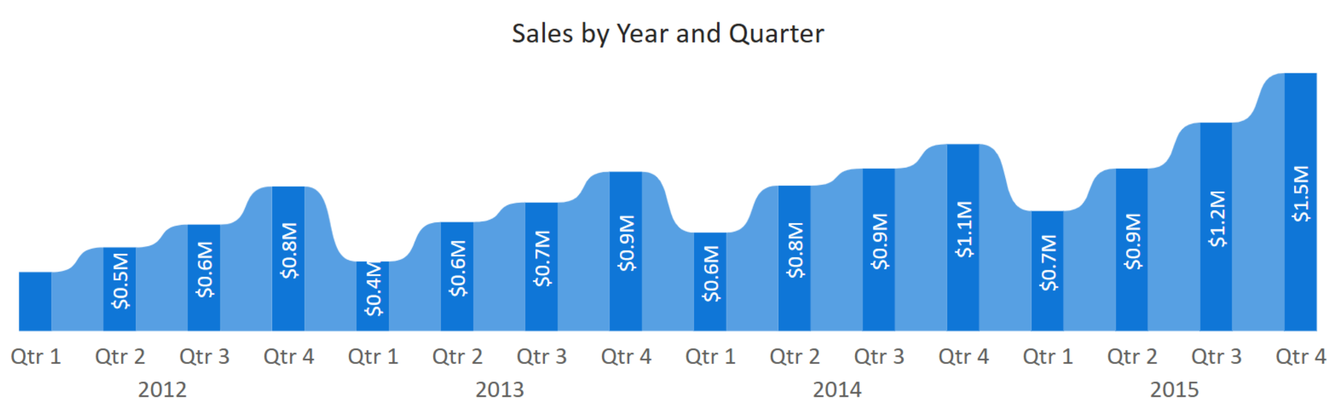
```
Lowest Sales Country =
VAR TopProduct =
    TOPN(1,
        SUMMARIZE('Fact_Sales', 'Dim_Geography'[Country], "Total_Sales", SUM(Fact_Sales[Sales Amount])),
        [Total_Sales],
        ASC
    )
RETURN
    CONCATENATEX(TopProduct, Dim_Geography[Country], ", ")
```

10) What is the trend of sales over years?



The company showed strong and consistent growth in sales from 2012 to 2015:

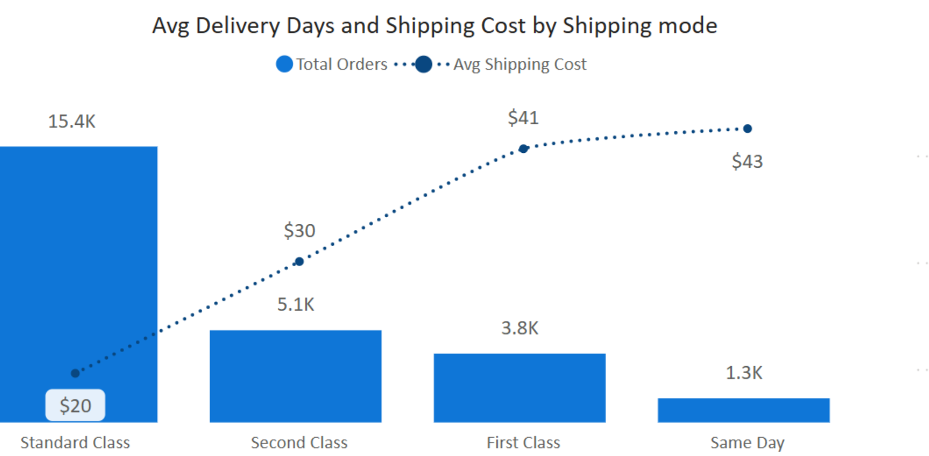
- 2012: $2.25 million (8,998 orders)
- 2013: $2.67 million (10,962 orders)
- 2014: $3.40 million (13,799 orders)
- 2015: $4.29 million (17,531 orders)



- Clear upward trajectory across all years
- Growth became more pronounced in later periods

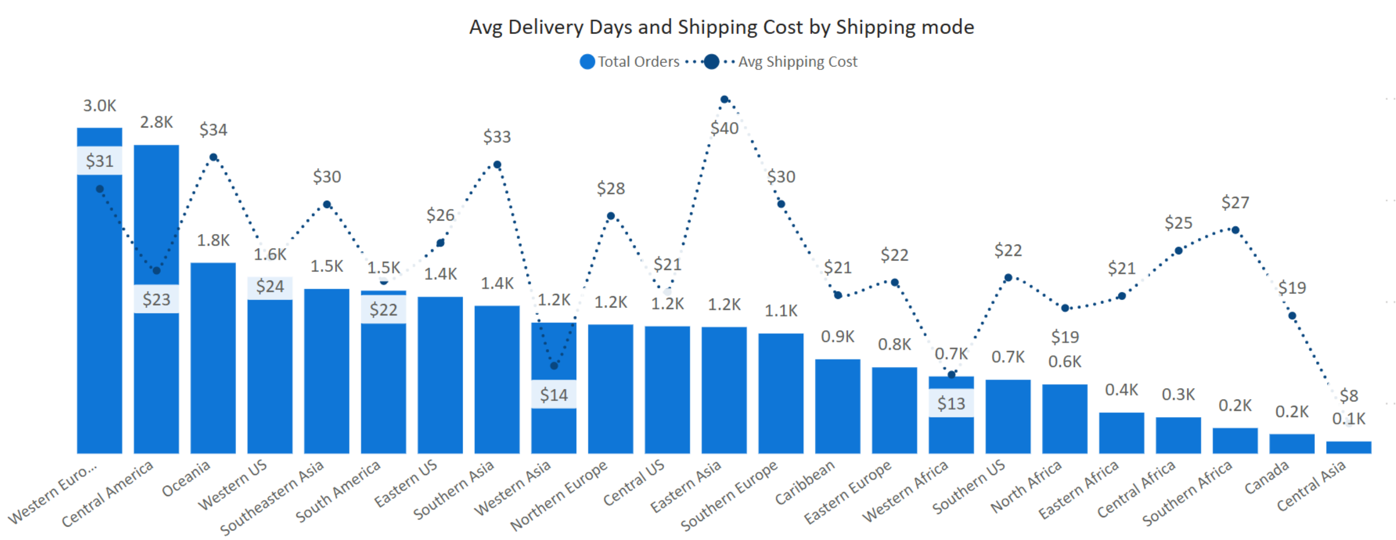
11) How can we reduce the shipping cost?

A) Optimize Ship Mode Selection



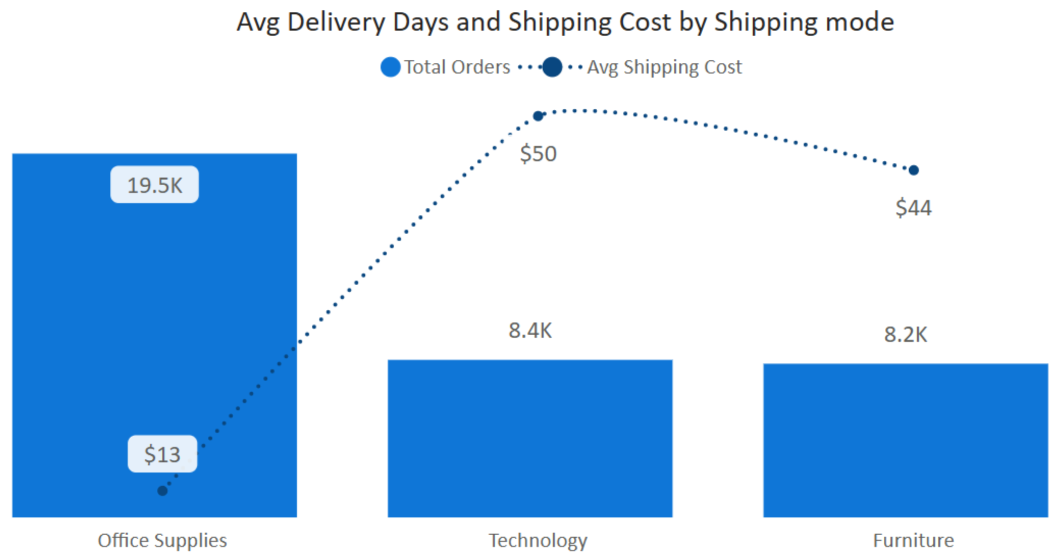
- Currently, Same Day shipping (avg cost: $43) and First Class (avg cost: $41) are significantly more expensive than Standard Class (avg cost: $20)
- The data shows that 1.3K orders used Same Day shipping with a shipping cost ratio of 17.25% of sales value.
- **Recommendation:** Limit Same Day and First-Class shipping to truly urgent orders only, as Standard Class is 54% cheaper and handles most orders (15.4k) efficiently.

B) Regional Optimization



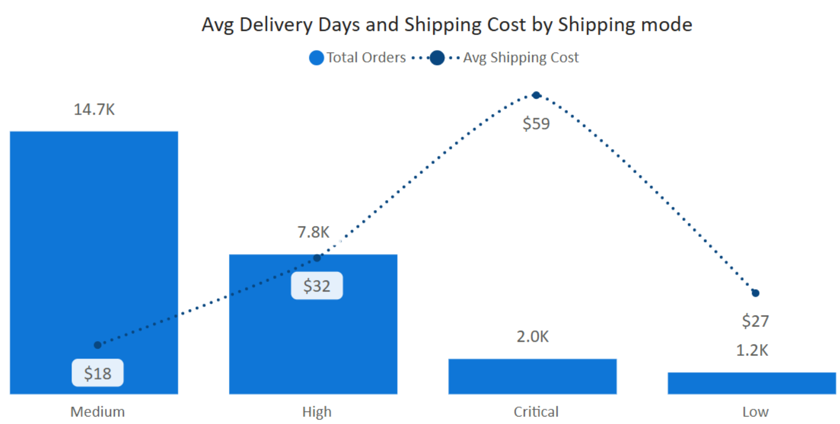
- Eastern Asia has the highest average shipping cost ($40)
- Central Asia has the lowest shipping cost ($8)
- Western Africa ($13) and Western Asia ($14) show much lower shipping costs
- **Recommendation:** Establish regional distribution centers in high-cost areas like Eastern Asia to reduce shipping costs for those 1.2K orders.

C) Product management optimization



- Technology items (avg shipping cost: ($50) and Furniture ($44) have significantly higher shipping costs than Office Supplies ($13)
- **Recommendation:** Bundle office supplies with furniture/technology orders since they have lower marginal shipping costs.

C) Priority Level Adjustment

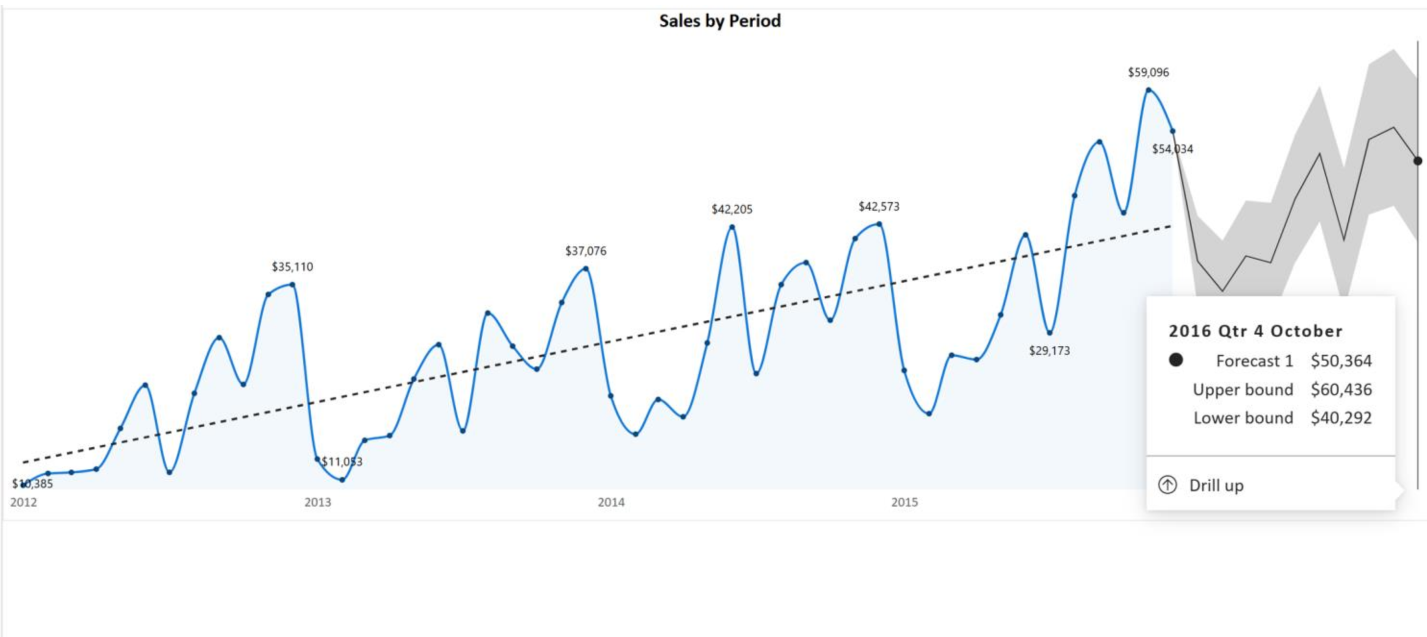


- Critical priority orders have the highest average shipping cost ($59)
- Medium priority orders have the lowest shipping cost ($18)
- **Recommendation:** Review the 2K critical priority orders and implement stricter criteria for critical shipping designation, as they cost 227% more than medium priority shipping.

12) How can we predict the future shipping costs?

We can do that in several ways either with Power BI or without it using other Statistical models.

1) Power BI Line charts



As we can see in this Line chart, we can predict the future shipping cost using **ETS model attached in Power BI line charts**, so we can see here that it predicts that the shipping cost for Oct. 2016 it will be **$60,436** and so on.

2) Statistical Models: -

- **Moving Average Model (MA):** This model calculates the average of past observations with the aim of predicting future values. It is useful for capturing short-term fluctuations and random variations in the data.
- **Autoregressive Model (AR):** This model predicts future values based on a linear combination of past observations.
- **Autoregressive Moving Average Model (ARMA):** The ARMA model combines the AR and MA models to capture both short-term and long-term patterns in the data. It is effective for analyzing stationary time series data.